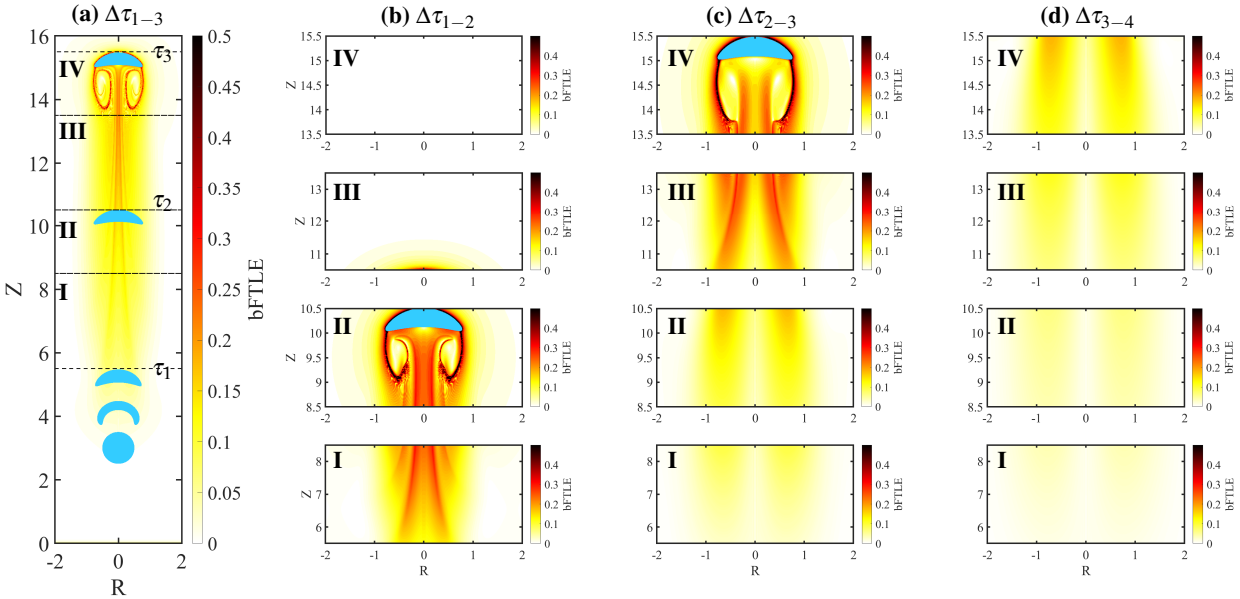

Figure 2: Backward FTLE fields of the rising bubble case C_6 (a) over the full interval $\Delta\tau_{1-3} = 4-20$, with observation windows indicated as I–IV. Temporal evolution of the FTLE fields within these windows, resolved into three equal sub-intervals: (b) $\Delta\tau_{1-2} = 4-8$ and (c) $\Delta\tau_{2-3} = 12-20$, and (d) $\Delta\tau_{3-4} = 20-28$, showing the temporal evolution of the FTLE field.